Advanced NumPy Proficiency Lab

Fancy Indexing, Broadcasting, Loss Functions & More

Based on: *2. Advance_numpy.pdf*

70 Hands-on Programming Questions (No Theory)

Solve by coding only — each question tests a specific NumPy concept.

BASIC: Array Creation & Properties

Question Create a 1D NumPy array of 10 elements and convert it to a Python list.

Question Create a 2D array of shape (3,4) and print its shape, size, dtype, and number of dimensions.

Question Compare the memory usage (in bytes) of a Python list of 1000 integers vs a NumPy array of 1000 integers.

Question Create a 2D array and use `reshape()` to change its shape to (2,6) and then flatten it.

Question Create a 5x5 identity matrix and extract the diagonal elements.

BASIC: Indexing & Slicing

Question Given `arr = np.array([10, 20, 30, 40, 50, 60, 70, 80])`, use fancy indexing to extract elements at positions `[1, 3, 5, 7]`.

Question Using fancy indexing, select elements from `arr` in reverse order `[7, 5, 3, 1]`.

Question Repeat indices `[0, 0, 2, 2, 4, 4]` from the array `arr = np.array([10, 20, 30, 40, 50])` using fancy indexing.

Question From a 2D array `arr_2d = np.array([[1,2,3,4],[5,6,7,8],[9,10,11,12]])`, use fancy indexing to select elements at rows `[0, 2]` and columns `[1, 3]`.

Question Given `arr = np.array([10, 15, 20, 25, 30, 35, 40])`, use boolean indexing to extract all elements greater than 25.

BASIC: Boolean Indexing

Question From the array `arr = np.array([10, 15, 20, 25, 30, 35, 40])`, extract elements between 20 and 35 (inclusive) using boolean indexing with `&` operator.

Question Extract elements less than 15 OR greater than 30 from the same array using `|` operator.

Question Given a 2D array `grades = np.array([[85,92,78],[65,70,95],[45,60,55]])`, replace all values less than 60 with 60.

Question From the same `grades` array, extract all scores greater than 80.

Question Create a boolean mask from `arr = np.array([1, 2, 3, 4, 5, 6])` that selects only even numbers.

INTERMEDIATE: Broadcasting

Question Add the scalar `10` to every element of `arr = np.array([1, 2, 3, 4, 5])` using broadcasting.

Question Create a 3x4 matrix and a 1x4 row vector. Add the vector to the matrix. Print the result and explain the broadcasting shape compatibility.

Question Create a 3x4 matrix and a 3x1 column vector. Add the vector to the matrix.

Question Create a column vector of shape (4,1) and a row vector of shape (3,). Add them to create a 4x3 grid. Print the shape and the result.

Question Determine if the following broadcasting operations are compatible:

(a) (3,4) + (4,)

(b) (3,4) + (3,1)

(c) (3,4) + (1,4)

(d) (3,4) + (3,5)

Question Normalize a dataset of shape (100, 10) by subtracting the mean (shape (10,)) and dividing by the standard deviation (shape (10,)) using broadcasting.

Question Given a 2D array of shape (5, 3), add a 1D array of shape (3,) to each row.

INTERMEDIATE: Mathematical Operations

Question Compute element-wise addition, subtraction, multiplication, and division of two arrays `a = np.array([10, 20, 30])` and `b = np.array([2, 4, 6])`.

Question Compute `a ** 2` and `a % b` for the same arrays.

Question Compute `sin(0)`, `sin(pi/2)`, and `sin(pi)` using NumPy.

Question Compute `cos(0)`, `cos(pi/2)`, and `cos(pi)`.

Question Compute `exp([1, 2, 3])` and `log([1, 2, 3])`.

Question Compute `log10([1, 10, 100])` and `log2([1, 2, 4])`.

Question Round the array `arr = np.array([3.14159, 2.71828, 1.41421, 2.5])` to 2 decimal places.

Question Compute `floor`, `ceil`, and `trunc` for the same array.

Question Compute `sqrt([1, 4, 9, 16])` and `square([1, 2, 3, 4])`.

INTERMEDIATE: Statistical & Aggregate Functions

Question For `matrix = np.array([[1,2,3],[4,5,6]])`, compute:

(a) Overall mean

(b) Mean along axis=0 (columns)

(c) Mean along axis=1 (rows)

Question Compute the sum, cumulative sum, standard deviation, and variance for the same matrix.

Question Find the min, max, median, and 25th percentile for the same matrix.

Question Compute the product and cumulative product for the same matrix.

Question For `a = np.array([1, 5, 3, 9, 2])` and `b = np.array([2, 4, 6, 8, 0])`, compute:

(a) Element-wise maximum

(b) Element-wise minimum

Question Clip the array `arr = np.array([1, 5, 10, 15, 20])` between 3 and 12.

Question Compute the absolute value and sign of `[-1, 0, 1, -2, 3]`.

INTERMEDIATE: Complex Numbers

Question Create a complex array `arr = np.array([1+2j, 3+4j, 5+6j])`.

Question Extract the real parts and imaginary parts of the array.

Question Compute the complex conjugate of the array.

Question Compute the absolute (magnitude) of the complex array.

INTERMEDIATE: Loss Functions

Question Implement the **Sigmoid function** $\sigma(x) = 1/(1+e^{-x})$ using NumPy and test it on $x = np.array([-2, -1, 0, 1, 2])$.

Question Verify the property $\text{sigmoid}(-x) = 1 - \text{sigmoid}(x)$ for $x = 2$.

Question Implement the **Mean Squared Error (MSE)** function: $\text{MSE} = \text{np.mean}((y\_true - y\_pred) * *2)$.

Question Test your MSE implementation with $y\_true = np.array([1, 2, 3, 4, 5])$ and $y\_pred = np.array([1.2, 1.8, 3.1, 3.9, 5.2])$.

Question Implement the **Binary Cross Entropy (BCE)** loss function: $\text{BCE} = -\text{np.mean}(y\_true * \log(y\_pred) + (1 - y\_true) * \log(1 - y\_pred))$.

Question Test your BCE implementation with $y\_true = np.array([1, 0, 1, 0, 1])$ and $y\_pred = np.array([0.9, 0.1, 0.8, 0.2, 0.95])$.

Question Test BCE with perfect predictions ($y\_pred = y\_true.astype(float)$) and observe the result.

ADVANCED: Handling Missing Values (NaN)

Question Create an array `arr = np.array([1, 2, np.nan, 4, 5, np.nan, 7])`.

Question Check for NaN values using `np.isnan()` and count how many NaNs exist.

Question Remove all NaN values from the array using boolean indexing.

Question Replace all NaN values with 0 using `np.nan_to_num()`.

Question Compute the mean of the array ignoring NaN values (`np.nanmean`).

Question Compute the standard deviation and sum ignoring NaN values.

Question Replace all NaN values with the mean of the non-NaN values using `np.where()`.

Question Differentiate between `None` and `np.nan` by:

(a) Creating arrays with each

(b) Checking their types

(c) Comparing them

(d) Creating a function that checks for missing values

ADVANCED: Visualization with Matplotlib

Question Create a simple line plot of $y = x^2$ for x from 0 to 10 with 20 points. Include labels, title, grid, and legend.

Question Plot $\sin(x)$ and $\cos(x)$ on the same figure for x from 0 to 2π with 100 points.

Question Create a 2x2 subplot showing:

(a) Linear ($y = x$)

(b) Quadratic ($y = x^2$)

(c) Sine ($y = \sin(x)$)

(d) Exponential ($y = e^x$)

Question Generate 1000 random numbers from a normal distribution and plot a histogram with 30 bins.

Question Create a bar chart with categories `['A', 'B', 'C', 'D', 'E']` and values `[25, 40, 30, 55, 45]` with different colors.

Question Generate 100 random points with $x \sim N(0, 1)$ and $y = 2x + 1 + 0.5 \cdot N(0, 1)$. Create a scatter plot with a regression line.

Question Customize a plot by:

(a) Setting linewidth to 2

(b) Adding fill between

(c) Customizing axis labels (fontsize, bold)

(d) Customizing ticks (0 to 10 with step 2)

(e) Adding a grid with dashed lines

(f) Adding an annotation at the maximum point

ADVANCED: Mixed Concepts

Question Write code to demonstrate that NumPy is faster than Python lists for element-wise operations (use `time` module).

Question Show that NumPy uses less memory than Python lists (use `sys.getsizeof()` and `nbytes`).

Question Given `arr = np.array([1, 2, np.nan, 4, np.nan, 6])`, create a function that detects and handles missing values (counts, removes, replaces).

Question Implement a function that takes a dataset of shape (N, D) and normalizes it using broadcasting (zero mean, unit variance).

Question Implement a function that computes both MSE and BCE for a given dataset and prints both.

Question Create a function that generates a dataset with noise and plots the data with a regression line.

Ready to Code!

"Broadcasting is your superpower — never write a Python loop for array operations again."

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